Day 38 EBIKE101 Diagnostic Worksheet

Program: CTS Cohort 6

Course: EBIKE101

Date: Tuesday, 2026-09-08

Project: E-bike Homunculous

Team members: _____

Purpose

Use this worksheet to diagnose what the Homunculous needs before it can be safely finished, powered, and eventually ridden.

Safety boundary

Do not connect the battery, bypass protection, open battery cases, test live battery circuits in continuity mode, spin the motor under power, or ride the bike without instructor approval.

1. Whole-bike inventory

Category	Present?	Secure?	Verified?	Notes / missing items
Frame / main structure				
Front wheel				
Rear wheel / motor wheel				
Tires / tubes				
Brakes				
Battery				
Battery mount				
Controller				
Controller mount				
Display				
Throttle				
PAS sensor / magnet ring				
Brake cutoff sensors				
Main fuse / protection				
Charger				
Labels / wire IDs				

2. System trace

Fill this out before powered testing.

System path	What part is it?	How do we know?	Still unknown
Source			
Protection			
Main switch/key path			
Controller power input			

Display/control input			
PAS input			
Throttle input			
Brake cutoff input			
Motor phase/hall cable			
Motor/load			

3. Symptom list

Symptom / issue observed	Possible causes	Test or inspection needed	Priority

4. Missing parts / supplies list

Needed item	Why it is needed	Required before power-up?	Who is checking?

5. Safe work approved today

Only list work the instructor has approved.

Task	Role / person	Safety concern	Done?	Instructor initials

6. Evidence log

Photo / note needed	Taken?	File name or location	Notes
Whole bike before work			
Battery/controller area			
Motor cable routing			
Missing parts			
Final state today			

7. Exit reflection

What is one thing we verified today?

_____

What is one thing we still do not know?

_____

What is the next safest work step?